Data Flow Diagram

Notary-Mediated Escrow System

System Architecture Documentation

Document Version 1.0
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1. Introduction

1.1. Purpose

This document provides a comprehensive Data Flow Diagram (DFD) for the Notary-Mediated Escrow System, illustrating how data flows through the system components.

1.2. Scope

The DFD covers:

- All smart contract interactions
- Frontend to blockchain communication
- User roles and permissions
- External system integrations

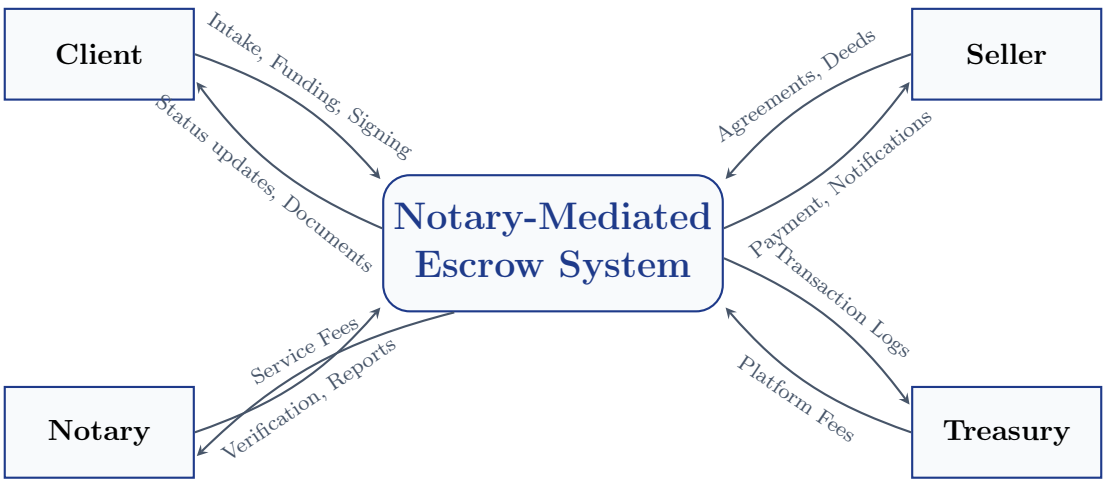
1.3. Definitions

- **Client:** Buyer/purchaser of property
- **Seller:** Property owner selling land
- **Notary:** Licensed professional verifying documents
- **Treasury:** Platform fee recipient
- **Multi-Sig Owners:** Signatories for fund releases

2. Context Diagram (Level 0)

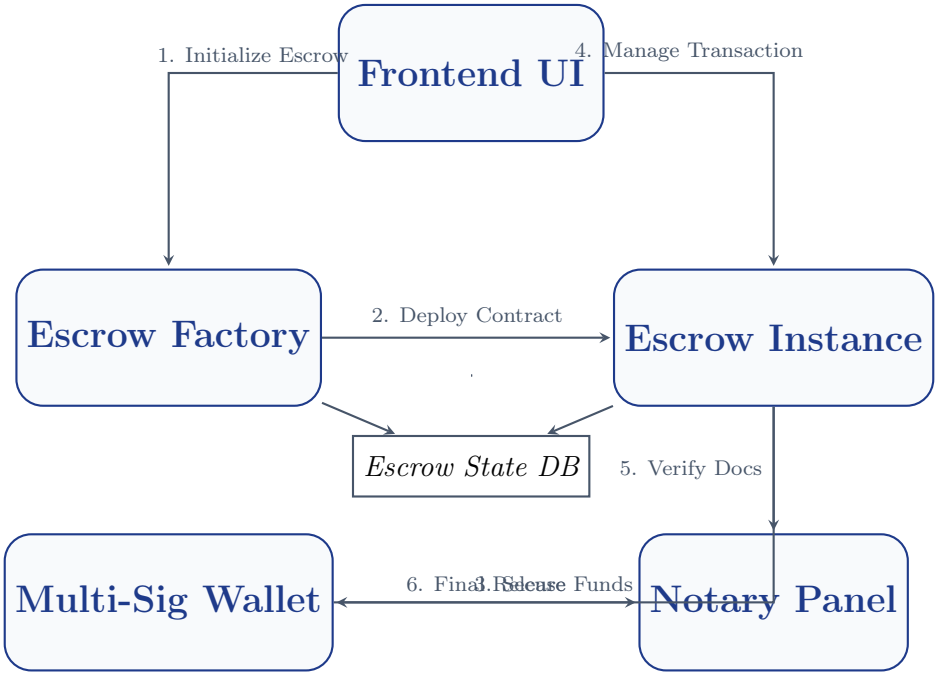
2.1. System Boundary

The system boundary encompasses all smart contracts and frontend components. External entities interact with the system through defined interfaces.



3. Level 1 DFD

3.1. Process Overview



3.2. Process 1: Client Intake



Data Elements:

- Input: budget, desiredSize, purpose, seller, notary
- Output: escrowId, created event
- Storage: EscrowTransaction struct

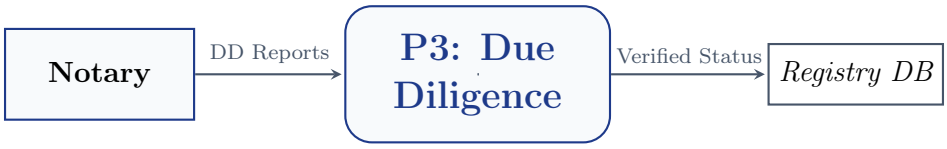
3.3. Process 2: Discovery Phase



Data Elements:

- Input: geoPins, photos, videos
- Output: gallery uploaded event

3.4. Process 3: Due Diligence



Documents:

- SearchCertificate
- DueDiligenceReport
- Site Visit Report

3.5. Process 4: Execution Phase



Flow:

1. Sign sale agreement
2. Notarize document
3. Client funds escrow
4. Execute fiat bridge
5. Disburse to seller

3.6. Process 5: Final Transfer



Documents:

- LodgementReceipt
- OriginalTitle
- LCBConsent

4. Data Stores

4.1. DS1: Escrow State

- Stores all escrow transactions
- Current state per escrow
- Milestone completion status

4.2. DS2: Documents

- Document hashes per escrow
- Verification status
- Signing status

4.3. DS3: Multi-Sig Wallet

- Owner list
- Pending transactions
- Confirmation status

5. External Interfaces

5.1. Ethereum Blockchain

- Smart contract deployment
- Transaction submission
- Event logging
- State queries

5.2. Frontend API

- Wallet connection (MetaMask)
- Contract interaction (ethers.js)
- Event
- State display

6. Flow Numbers

6.1. P1: Client Intake

1. Client submits intake data
2. Contract validates budget $\geq$ due diligence fee
3. Escrow created with new ID
4. State set to Intake

6.2. P2: Property Discovery

1. Notary uploads digital gallery
2. Escrow state transitions to Discovery
3. Client reviews properties
4. Client selects plot
5. State transitions to PlotSelected

6.3. P3: Due Diligence

1. Client funds escrow with due diligence fee
2. State transitions to DueDiligencePending
3. Notary uploads DD report
4. Client verifies documents
5. State transitions to DueDiligenceComplete
6. Due diligence fee released to notary

6.4. P4: Execution

1. Client proceeds with purchase
2. Both parties sign agreement
3. Notary notarizes document
4. Client funds remaining amount
5. Notary executes fiat bridge
6. State transitions to ExecutionComplete
7. Deposit released to seller

6.5. P5: Final Transfer

1. Notary uploads lodgement receipt
2. State transitions to TransferRegistry
3. Notary confirms title documents
4. Multi-sig confirms final release
5. Facilitation fee to treasury
6. Balance to seller
7. State transitions to Completed

7. Summary

This DFD illustrates the complete data flow through the Notary-Mediated Escrow System. Key observations:

1. State machine provides strict phase enforcement
2. Multi-sig protects all fund movements
3. Notary acts as trusted oracle
4. Documents hashed for immutability
5. Fiat bridge enables local currency conversion

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